

Summer Internship Experience – Dover

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About Dover

Dover is a diversified global manufacturer and solutions provider headquartered in Downers Grove, Illinois, USA, with revenue exceeding \$7 billion. The company provides innovative equipment and components, consumable supplies, aftermarket parts, software and digital solutions, and support services across five operating segments: Engineered Products, Clean Energy & Fueling, Imaging & Identification, Pumps & Process Solutions, and Climate & Sustainability Technologies. For over 70 years, Dover has been recognized for its entrepreneurial approach. With a team of approximately 24,000 employees, the company embraces an ownership mindset and works closely with customers to redefine what's possible.

Dover visited our campus for summer internship hiring. The usual selection process included one online assessment, two technical interviews, and an HR round. However, in my case, I had only one online assessment followed by a single technical interview.

Round 1: Online Assessment

The test consisted of 60 multiple-choice questions to be completed in 60 minutes. It was divided into four sections:

- Computer Science Fundamentals (15 questions)
- Logical Reasoning (15 questions)
- Quantitative Aptitude (15 questions)
- Data Structures and Algorithms (15 questions)

The level of difficulty was moderate, but speed and accuracy were essential. With consistent practice in problem-solving platforms, I found this round manageable. Out of all the participants, 15 students were shortlisted for the next round.

Round 2: Technical Interview

Since I had highlighted my strength in problem-solving, the interview mainly focused on DSA and project discussion.

- **DSA Problems:**

- **Count Number of Nice Subarrays:**
The interviewer asked me to solve this problem, as it was one of the daily challenges I had recently solved on LeetCode. I walked them through the problem, explained the naive and optimized approach, and wrote clean code using prefix sums and hashmap for an efficient solution.
- **Frequency of the Most Frequent Element:**
For this, I was asked to explain only the logic. I discussed the sliding window technique after sorting, maintaining a window with two pointers, and how to keep track of the current frequency. I also mentioned the time complexity and edge cases.
- **Project Discussion:**
I was asked to explain one of the projects I had listed on my resume. I chose a full-stack project where I used DSA in backend logic. I explained the overall architecture, the flow between frontend and backend, and how specific data structures (like hashmaps and arrays) helped in efficient data handling.
The interviewer also asked why I chose certain structures and whether the system could scale with larger data.

Overall, it was a very engaging and focused interview. I appreciated that the interviewer took interest in my LeetCode profile and asked relevant questions.

Out of 15 shortlisted students, only 3 were selected for the summer internship at Dover, and I was fortunate to be one of them.

Tips for Preparing for Dover Internship

1. **Master the Basics + DSA:**
Focus on CS fundamentals (OS, DBMS, CN, OOP) and practice DSA daily. Most interview questions come from topics like sliding window, prefix sum, and hashmaps.
2. **Practice Aptitude & Logical Reasoning:**
The online assessment isn't just coding—quantitative and logical questions also matter. Practice regularly on platforms like IndiaBix and GeeksforGeeks.
3. **Keep Your Profiles Ready:**
Update your LeetCode and GitHub profiles. Interviewers might ask to see your recent activity or pick questions based on your solved problems.
4. **Know Your Projects Well:**
Be ready to explain your projects, especially the logic and DSA used. Highlight design decisions, challenges faced, and how you solved them.

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